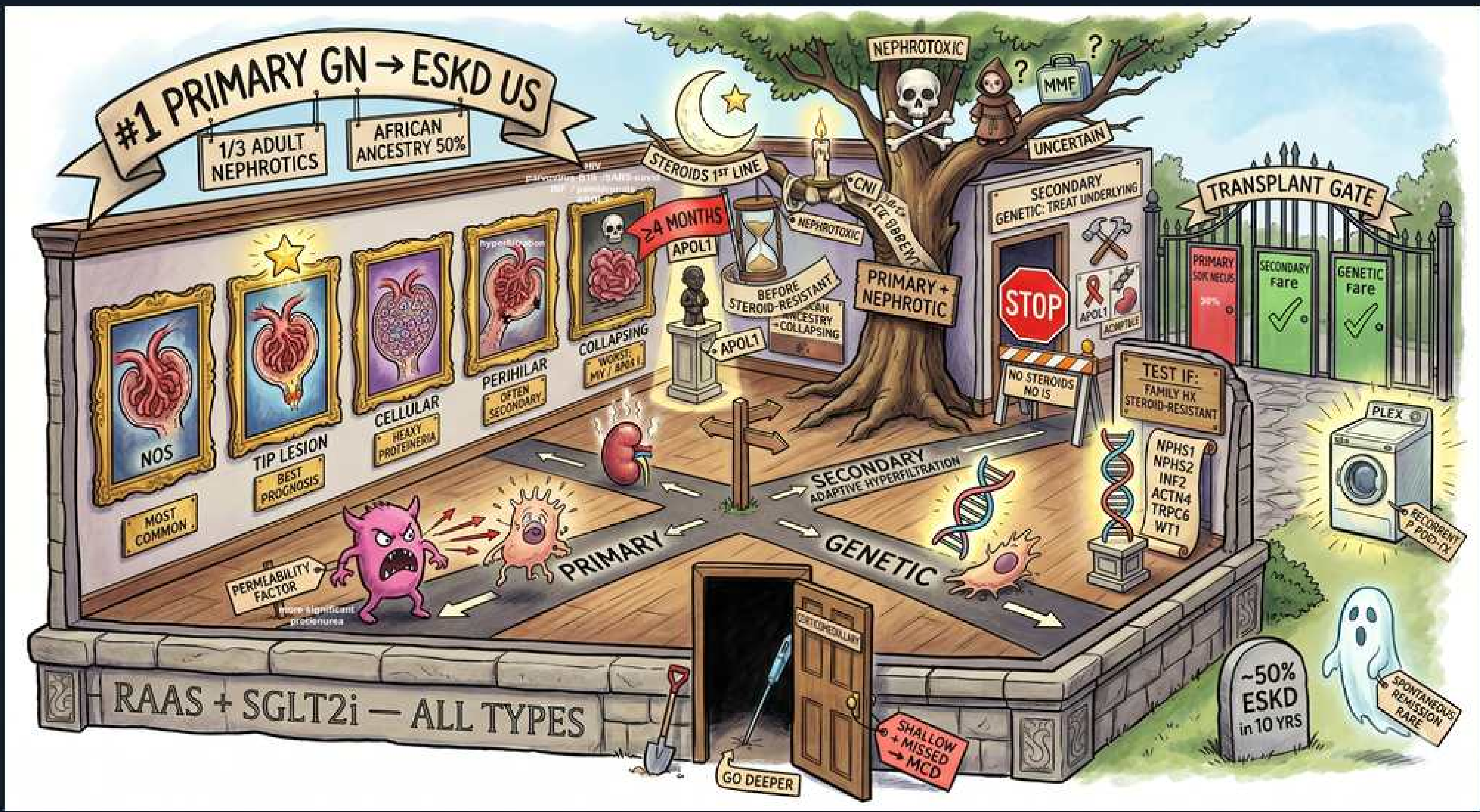
Focal Segmental Glomerulosclerosis (FSGS) — #1 Primary GN to ESKD



1. #1 primary GN → ESKD (US)

WHAT YOU SEE

Banner: #1 PRIMARY GN → ESKD US

WHAT IT ENCODES

FSGS is the most common primary glomerular cause of nephrotic syndrome AND of ESKD among primary GNs in the US. ~1/3 of adult nephrotic syndrome. Disproportionately affects patients of African ancestry (~50%), linked to APOL1 risk alleles. Board point: nephrotic-range proteinuria + a sclerotic segment of some (focal) glomeruli on biopsy.

BOARD TRAP

Membranous is the classic 'most common primary nephrotic in white adults' — but FSGS leads overall and drives progression to ESKD.

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2. 1/3 adult nephrotics

WHAT YOU SEE

Scroll: 1/3 ADULT NEPHROTICS

WHAT IT ENCODES

FSGS accounts for roughly one-third of nephrotic syndrome in adults and is rising in incidence. Presents with edema, heavy proteinuria, sometimes hematuria and HTN. Often subnephrotic at presentation, unlike MCD which is abruptly full-blown nephrotic.

BOARD TRAP

MCD is the #1 cause in CHILDREN; FSGS is the rising adult cause and the one that scars.

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3. African ancestry ~50% / APOL1

WHAT YOU SEE

Banner: AFRICAN ANCESTRY 50%; APOL1 hourglasses

WHAT IT ENCODES

Two APOL1 risk variants (G1/G2) dramatically increase risk of FSGS, HIVAN, and hypertensive nephrosclerosis in people of West-African descent. APOL1 high-risk genotype predicts faster progression and worse transplant outcomes.

BOARD TRAP

APOL1 explains African-ancestry FSGS/HIVAN risk — not a complement or anti-PLA2R mechanism.

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4. Histologic variants (NOS/tip/cellular/perihilar/collapsing)

WHAT YOU SEE

Framed glomeruli: NOS, TIP LESION, CELLULAR, PERIHILAR, COLLAPSING

WHAT IT ENCODES

Columbia classification: NOS (most common), tip lesion (best prognosis, steroid-responsive), cellular, perihilar (seen in secondary/adaptive FSGS), and collapsing (worst prognosis, associated with HIV/APOL1/pamidronate). Light microscopy shows segmental sclerosis; the rest of that glomerulus and many others look normal — sampling matters.

BOARD TRAP

Tip lesion = best outcome; collapsing = worst. Don't reverse them.

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5. Foot process effacement (EM)

WHAT YOU SEE

Lower-left: MICRO HEMATURIA / podocyte cartoon

WHAT IT ENCODES

EM shows diffuse podocyte foot process effacement — shared with MCD. The discriminator is segmental sclerosis with hyalinosis on LM and a permeability/podocyte injury 'factor' in primary disease. IF is negative or shows nonspecific IgM/C3 trapping in sclerotic segments.

BOARD TRAP

Effacement alone can't separate FSGS from MCD; FSGS adds focal segmental scarring on LM.

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6. Permeability factor / podocyte injury

WHAT YOU SEE

Pink monster: PERMEABILITY FACTOR

WHAT IT ENCODES

Primary FSGS is driven by a circulating permeability factor (e.g., suPAR candidate) injuring podocytes — explains rapid recurrence in transplants, sometimes within hours. Plasmapheresis can treat post-transplant recurrence.

BOARD TRAP

Rapid post-transplant recurrence with massive proteinuria = primary FSGS permeability factor, not rejection.

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7. Primary vs secondary vs genetic

WHAT YOU SEE

Floor compass: PRIMARY / SECONDARY / GENETIC

WHAT IT ENCODES

Classify FSGS before treating. Primary (immune/permeability factor) → immunosuppression. Secondary (adaptive: obesity, reflux, reduced nephron mass, HTN, anabolic steroids) → RAAS blockade, treat cause. Genetic (NPHS1/NPHS2/podocin, etc.) → generally steroid-resistant.

BOARD TRAP

Steroids help PRIMARY FSGS; in secondary/genetic FSGS steroids are ineffective and may harm — use RAAS/SGLT2 instead.

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8. Primary nephrotic → STOP / treat

WHAT YOU SEE

Tree, STOP sign, PRIMARY NEPHROTIC

WHAT IT ENCODES

Primary FSGS with full nephrotic syndrome is treated with high-dose corticosteroids for an extended course (often months), then calcineurin inhibitors if steroid-resistant. Heavy proteinuria predicts progression; spontaneous remission is uncommon.

BOARD TRAP

Long steroid courses are for primary FSGS only; never immunosuppress purely adaptive/secondary FSGS.

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9. Steroids ~6 months

WHAT YOU SEE

Hourglass: 6 MONTHS, STEROIDS

WHAT IT ENCODES

Primary FSGS often needs prolonged (3–6+ month) steroid therapy to achieve remission — slower response than MCD. Steroid-resistant disease moves to CNI (cyclosporine/tacrolimus).

BOARD TRAP

FSGS responds slower and less completely than MCD; don't expect MCD-like rapid remission.

10. Secondary → treat underlying

WHAT YOU SEE

Right banner: SECONDARY — TREAT UNDERLYING

WHAT IT ENCODES

For secondary FSGS, target the driver: weight loss/SGLT2 for obesity-related, correct reflux, stop anabolic steroids, control HTN. ACEi/ARB to reduce intraglomerular pressure and proteinuria. No immunosuppression.

BOARD TRAP

Immunosuppressing adaptive FSGS is a classic wrong answer — RAAS blockade is the move.

11. Genetic FSGS / podocin

WHAT YOU SEE

Right: GENETIC, DNA, podocyte genes

WHAT IT ENCODES

Inherited podocytopathies (NPHS1 nephrin, NPHS2 podocin, WT1, INF2, ACTN4, TRPC6) cause steroid-resistant FSGS, often pediatric. Genetic testing guides avoiding futile immunosuppression and informs transplant counseling (low recurrence risk).

BOARD TRAP

Genetic FSGS is steroid-RESISTANT and has LOW transplant recurrence — opposite of primary permeability-factor FSGS.

12. Transplant recurrence risk

WHAT YOU SEE

TRANSPLANT GATE, washing machine

WHAT IT ENCODES

Primary FSGS recurs in ~30–40% of allografts (sometimes immediately) due to circulating factor; treated with plasmapheresis ± rituximab. Genetic/secondary FSGS rarely recurs.

BOARD TRAP

High recurrence = primary; near-zero recurrence = genetic. Match the mechanism.

13. RAAS + SGLT2i — all types

WHAT YOU SEE

Bottom banner: RAAS + SGLT2i — ALL TYPES

WHAT IT ENCODES

Regardless of subtype, ACEi/ARB plus SGLT2 inhibitors are foundational antiproteinuric/renoprotective therapy for FSGS. They lower intraglomerular pressure and slow progression to ESKD.

BOARD TRAP

RAAS/SGLT2i is universal background therapy; it does not replace immunosuppression in true primary FSGS.

14. ~50% ESKD in 10 yrs

WHAT YOU SEE

Ghost circle: -50% ESKD IN 10 YRS

WHAT IT ENCODES

Untreated/refractory primary FSGS progresses to ESKD in roughly half of patients within ~10 years; complete remission of proteinuria is the strongest predictor of renal survival.

BOARD TRAP

Achieving remission — not just lowering proteinuria partially — is what protects the kidney long-term.